Diabetes Patient Dataset

Data cleaning and exploratory data analysis report | 2 November 2026

Encounters	Joined columns	Cleaned splits	Join key
101,766	56	6	encounter_id

Purpose and result

The original wide diabetes encounter file was divided into six manageable CSV files. Each split was cleaned and explored separately, then the six cleaned outputs were joined using the shared encounter_id field. The final file contains one row per encounter and retains the split-level structure for review.

What the EDA covered

• Initial inspection: dataframe preview, schema and data types, dimensions, descriptive statistics, missing-value counts, duplicate-row checks, duplicate-key checks, and distinct-value counts.

• Feature exploration: numeric summaries, histograms and box plots; categorical frequency tables and top-category bar charts. Diagnosis codes were treated as categories rather than numeric measurements.

• Relationship exploration: numeric correlation heatmaps in the splits with multiple numeric measures, plus comparisons of diagnosis counts and prior outpatient visits across readmission groups in dataset 6.

• Output checks: row and key preservation, age-bound checks, column uniqueness, missing-value review, and validation that each saved CSV could be reloaded.

Cleaning decisions

Decision	What was done
Missing markers	Question-mark and blank text values were interpreted as missing. For diagnosis, payer, and race fields, a missingness flag was retained and the display category was labeled Unknown. For glucose and A1C test results, missing values were labeled Not measured, with a separate missingness flag.
Age interval	The age band was parsed into age_min and age_max. The original age field was removed after parsing; bounds were checked for completeness and order.
Weight	Removed from dataset 2 because 98,569 of 101,766 values were missing (96.86%).
Constant features	citoglipton (dataset 5) and examide (dataset 6) each had only the value No and were removed because they have zero variation.
Duplicates and rows	No full duplicate rows or duplicate encounter_id values were found in the source splits. No encounter rows were removed.
Rare categories	Highly imbalanced but non-constant medication fields were retained. Their rare observed values may contain useful information for later analysis or modeling.

Cleaning by split

Split	Fields and treatment	Final dimensions
Dataset 1	Primary diagnosis missingness flag; diagnosis code kept categorical.	101,766 × 11
Dataset 2	weight removed; age expanded to age_min and age_max; third-diagnosis missingness flag.	101,766 × 10
Dataset 3	payer_code missingness flag and Unknown label.	101,766 × 10
Dataset 4	race missingness flag and Unknown label; patient_nbr preserved.	101,766 × 10
Dataset 5	missingness flags for max_glu_serum, A1Cresult, medical_specialty; missing tests labeled Not measured; gender normalized; citoglipton removed.	101,766 × 11
Dataset 6	secondary diagnosis missingness flag; examide removed; readmission status retained.	101,766 × 9

Final joined file checks

Check	Result
Shape	101,766 rows × 56 columns
Join key	encounter_id is present, unique, and aligned across all six cleaned splits. One-to-one inner joins preserved all encounters.
Remaining missing cells	0
Constant non-key columns	[]
Unresolved question-mark markers	None found in the cleaned dataset.

Interpretation and limits

The diagnosis fields contain ICD-style code strings. This EDA reports their frequency as codes; it does not add a diagnosis-description codebook. The age columns represent interval bounds, not exact ages. Correlations and readmission group comparisons are exploratory associations and do not establish causation or clinical effects. No clinical outliers were automatically deleted.

Files produced

Six cleaned CSV files are stored in EDA_Dataset, alongside the joined EDA_Full_Dataset.csv. Six executed notebooks document each split, and EDA_Full_Dataset_assembly.ipynb documents the join and its checks. Each notebook has inline comments beside executable code.

Final dataset path: EDA_Dataset/EDA_Full_Dataset.csv